*Millor*IA

Graduation Internship

A memory-based cognitive stimulation platform.

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Colophon

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Introduction

About the project

This project is an AI-powered platform called MillorIA. It is designed to support individuals with Mild Cognitive Impairment (MCI). MCI leads to gradual declines in memory, attention, language and executive function. The platform aims to help by slowing down or stabilizing progression in symptoms.

The website serves as an interactive training tool that can be used collaboratively by the person with MCI and their family members or caregivers. Users can upload photos, audio recordings, and text to create meaningful "memory collections." Using AI, the platform then generates custom exercises designed to stimulate memory recall, support emotional connection and promote cognitive engagement in a familiar and comforting way. This is inspired by a form of therapy called reminiscence therapy. Exercises range from multiple choice to inserting the right answer or putting events in the right order. Unlike existing cognitive training apps, which often offer generic exercises, this platform works with personal memories.

About the project

VRAIN, Valencian Research Institute for Artificial Intelligence, is an AI research lab based in the province of Valencia, Spain, and is part of the Universitat Politècnica de València. The institute brings together researchers from eight different research groups, including areas such as language engineering, machine learning, formal languages, logic programming, software production and interactive technologies. VRAIN has over 30 years of experience in artificial intelligence, and has become a leading research center both nationally and internationally. The institute is home to 176 researchers, making it one of the largest AI research centers in Spain. VRAIN's work focuses on applying AI to real-world challenges in a wide range of fields, including healthcare, agriculture, industry, security, robotics, energy, and environmental sustainability. Their projects aim to develop innovative and impactful solutions through cutting-edge research.

The subdivision where I will be working is named Vertexlit and is based on the campus of Alcoi, also part of the Universitat Politècnica de València. Existing of a small team of researchers and phd students.

Project Overview

My role in the project

In this project, my role is to design and develop the user interface part of the platform. I am responsible for translating my research insights into a clear, accessible design and developing this into a functional platform. This includes researching the design requirements, producing wireframes and prototypes, conducting tests with the target group and implementing the final interface in React. I ensure that the platform is intuitive to use, accessible, visually consistent and technically functional. Although I integrate the database and the AI model to support user interactions, my main focus in this paper remains on the frontend experience rather than the backend systems.

Scope and boundaries

What I will do

- Conduct research into the target group.
- Create wireframes and a high-fidelity prototype.
- Ensure that the interface is accessible for people with Mild Cognitive Impairments.
- Test the interface with representative testers and refine the design based on feedback.
- Design and build the user interface and the platform. This includes the backend integrations of the database and AI model.

What I won't do

- **I will not develop the AI models.** The focus is on the interface; The AI model falls outside the scope and will be delivered by Isabel Mollá. I can not control the outcome or the quality of the exercises outside of providing a user-friendly interface for them.
- **I will not create medical or a clinical invention.**The platform may be used to support MCI, but does not replace professional treatment. Although this platform may have a positive effect on people I will not advertise it as a medical invention. This is to avoid the risk that users may view the platform as a replacement for professional treatment and for that reason not seek out professional treatment. This platform is not based on enough clinical research to replace any legitimate form of treatment.
- **I will not solve all accessibility challenges for all impairments.** This research does not aim to solve all accessibility challenges for every type of impairment. Its focus is specifically on people with Mild Cognitive Impairment (MCI), rather than on all possible disabilities. That said, the platform will comply with basic accessibility standards, meaning it remains accessible to a broad audience larger than the primary target group.
- **I will not conduct large-scale clinical testing.** Any user tests conducted are small and solely focused on usability and accessibility.

Definition of done

MillorIA is considered done when the platform has a carefully thought-out, accessibility-focused design that is based on research into the needs of the target group. It has been developed in line with accessibility standards and implemented as a working prototype with a scalable database and an integrated AI model.

The project will not be published on a public domain, as the AI model still requires further iteration before it would be suitable for real-world use. In the definition of done stage, MillorIA serves as a solid and fully functioning prototype of an accessible, memory-based cognitive stimulation platform.

Problem Definition

Research questions

Main research question

How can I design and develop a user-friendly interface for a cognitive-stimulation platform that helps older adults with mild cognitive impairments?

Sub-research questions

- Why is it important to have the internet be accessible for older adults with mild cognitive impairments?
- What is mild cognitive impairment and how does it affect the way patients interact with digital interfaces?
- Which design principles and interaction patterns are most effective for making interfaces accessible and intuitive for this target group?
- What are the needs of older adults with mild cognitive impairment when interacting with digital interfaces?

Why accessibility for cognitive impairments matters

The global population is getting older. According to the U.S. Census Bureau, by 2050 there will be a much bigger increase in older adults than in younger people and the old and young balance will have shifted (He & Goodkind, 2019). That means that the digital landscape will be changing as well. The future seniors will bring a lifetime of internet experience with them, continuing to engage with digital tools and online marketplaces far more naturally than the current generation.

At the same time, cognitive impairments are expected to become significantly more common as populations age. The World Health Organization reported that in 2015 approximately 47 million people worldwide were living with dementia, and this number is predicted to almost triple by 2050. Dementia and other cognitive disorders are already among the leading causes of disability and dependency in older adults (WHO, 2023). Even outside of dementia, mild cognitive impairment (MCI) affects a substantial group of older adults, influencing memory, attention, and processing speed.

This all means that in the future, more online users will be older adults and many of them may have some level of cognitive difficulty. Because of this, designing websites and digital products with this group in mind is important both ethically and economically. Ethically, everyone should be able to access and use digital services. Economically, older adults form a growing group of online consumers with strong purchasing power. If a website is hard to use, companies risk losing these customers.

That is why it is important that this group is not overlooked when it comes to internet use. As the number of older adults and people with cognitive difficulties increases, accessible and easy-to-use websites should become a standard rather than an exception.

Theoretical Research

Research methods

This project has multiple forms of research. A literature study was used to gain theoretical insight into MCI, age-related cognitive and physical changes and accessibility guidelines. In addition, an expert interview with a neurologist was used to compare these findings with real-world experience. Finally, iterative usability testing with the target group was conducted to refine the interface based on user behaviours and feedback.

What is mild cognitive impairment?

Cognitive decline is a very normal part of aging. MCI is diagnosed when these changes become interrupting in daily life, but do not yet prevent the person from carrying out their usual activities. For most people MCI represents a transitional stage between healthy aging and dementia and affects 10–15% of the population over the age of 65 (Anderson, 2019).

People with MCI will have problems with memory, language, attention difficulties and possible problems with executive functioning. MCI can be an early stage of dementia, although some studies indicate that it does not always progress to dementia (Mayo Clinic Staff, 2024). While dementia is always progressive, this is not necessarily the case for MCI. However, in an interview with José Manuel Moltó Jordá, a neurologist with extensive experience treating Alzheimer's patients, he noted that in his experience, patients with MCI almost always deteriorated over time and eventually developed dementia.

Symptoms of MCI

- memory or learning – difficulties remembering recent events or learning new things
- reasoning – struggling to make decisions or work through everyday problems
- attention – finding it more challenging to focus on a task or filter out distractions
- language – having difficulties finding the right word in conversation
- loss of interest or motivation – less interest in usual activities or hobbies.

User and Design Research

Target group

The platform is designed for people with Mild Cognitive Impairment (MCI). This includes mainly older adults (65+) with memory or thinking difficulties. In addition, caregivers and family members will be involved in supporting or guiding the use of the platform.

Older adults diagnosed with MCI often experience additional age-related impairments, such as changes in vision, hearing, motor skills, and digital confidence. Combined with the fact that the symptoms and progression of MCI differ per individual, this means the target group cannot be considered homogeneous. For this reason, the research also addresses general accessibility challenges faced by older adults using the internet, on top of those specifically related to MCI.

Designing for seniors

The internet isn't always the most accessible place for everybody. But when is something accessible? The United Nations state that accessibility refers to the practice of designing products, services and environments so that they can be easily used by all people, regardless of their age, abilities or disabilities, ensuring that everyone can participate equally. The most recognized standard for digital accessibility on the internet is the Web Content Accessibility Guidelines (WCAG) 2.1 developed by the World Wide Web Consortium (W3C). WCAG are a set of international rules set to make the internet more accessible especially for people with disabilities. The WCAG has three different success criteria, A, AA and AAA, going from least strict to most strict. Impairments the WCAG covers include vision, physical ability, hearing and cognitive ability.

There are no separate accessibility guidelines specifically for older adults. However stated by the World Wide Web Consortium (W3C) on their page about accessibility issues for older adults:

"These issues overlap with the accessibility needs of people with disabilities. Thus, websites, applications, and tools that are accessible to people with disabilities are more accessible to older users as well." - (W3C, 2025)

However, WCAG compliance alone is insufficient according to T. Gilbertson, 2015. Accessibility guidelines can for example not address a lack of confidence in using technology by older people. Research has shown that the majority of ICT using seniors feel "intimidated" and "anxious" about using technology and anticipate that the Internet is difficult to use and to understand. This fear of making mistakes can limit their willingness to explore, learn, and build digital skills (Marquié et al. 2002; Turner et al. 2007).

More importantly, WCAG does not consider that older adults differ from people with disabilities in how they use technology. Many older users are not even aware of the assistive technologies embedded in operating systems and browsers that could help them navigate the web. And without this awareness, they cannot take advantage of the built-in support designed to improve accessibility. (Wu, S et al 2021)

User Challenges and Accessibility Implications

From user challenges to design decisions

The below descriped age-related and cognitive impairment related changes explain why specific accessibility choices are necessary. The guidelines in the later chapter "Accessibility guidelines for MillorIA" translate these physical and cognitive limitations into concrete interface decisions, such as color contrast, text size and interaction design. These age-related and cognitive impairment related changes explain why specific accessibility choices are necessary. The guidelines in the later chapter "Accessibility guidelines for MillorIA" translate these physical and cognitive limitations into concrete interface decisions, such as color contrast, text size and interaction design.

Age related challenges

Vision

As people age, their ability to perceive contrast decreases, both because the eye receives less light and because neural processing becomes less efficient. Research shows that contrast sensitivity declines a lot in later adulthood, making it harder to detect low- or medium-contrast elements (Owsley, Sekuler & Siemsen, 1983). Additional studies demonstrate that age-related optical changes, such as yellowing of the lens and reduced retinal illumination, increase visual "noise," making fine detail and subtle color differences more difficult to see. These findings show that low-contrast interfaces create an additional barrier for older users, which has direct consequences for color design choices. Research also suggests that extremely bright white backgrounds can cause eye-fatigue so interfaces designed for older adults benefit from high color contrast but should avoid glaring white backgrounds (Owsley, 2011).

Hearing

Hearing loss is one of the most common age-related impairments and it has direct links to cognitive decline. Large studies show that hearing impairment is associated with faster declines in memory, attention and cognition (Lin et al., 2020). Other research indicates that untreated hearing loss increases the risk of dementia, partially because of reduced communication and social isolation (Deal et al., 2017). Since online content increasingly relies on audio and video, designers should ensure clear transcripts and captions to reduce barriers for older users.

Motor Skills

Physical changes such as reduced hand strength, tremors, arthritis or slower movement can make precise mouse or touch interactions more difficult for older adults. Studies show that older users (aged 60–75) perform significantly worse on tasks requiring double-clicking, dragging or rapid cursor movement compared to younger users (Keates & Trewin, 2005).

Cognitive Impairment

Normal aging leads to slower processing speed, reduced working memory and decreased attentional capacity. Research shows that older adults experience progressive declines in recall and processing speed (Salthouse, 1996). Other studies on aging also show that reductions in working memory and processing speed are strongly tied to cognitive decline over time (Caspi, Zimprich & Spitzer, 2022). These changes result in older adults often struggling with multi-step tasks, inconsistent navigation or dense information. This means that interfaces for this target group must be predictable and linear, so that users are not confronted with multiple choices or unexpected steps at the same time.

Mild cognitive impairment related challenges

Task performance and cognitive load

Individuals with MCI can face significant challenges when using digital interfaces. Research shows that people with MCI take longer to complete tasks, make more mistakes, and struggle more with complex or multi-step processes compared to older adults without cognitive impairments. Cognitive difficulties such as impaired planning, judgment, problem-solving and decision-making also make it harder to locate information, integrate different pieces of information and act on it effectively (Jokisuu et al., 2011). Time-limited tasks should be avoided or made adjustable, as users with MCI need more time to read, interpret and respond to information (Haesner et al., 2017).

Navigation and orientation

Eye-tracking studies indicate that they shift their gaze around the screen more frequently, suggesting uncertainty about where to click or what action to take next (Haesner et al., 2017). Users may lose track of where they are within a process, forget previous steps, or become disoriented when too many actions or navigation layers are available at once. Interfaces must therefore support orientation, predictability and step-by-step progression.
Research suggests that linear navigation, where users follow a guided sequence of steps, reduces disorientation and cognitive load, making digital tasks more manageable (Castilla et al., 2018).

Authentication and memory load

Authentication steps often create friction: remembering passwords, managing two-factor authentication codes, or handling CAPTCHAs can be overwhelming. To support users, the platform should minimize the need for memory recall.

Visual load and layout complexity

Visual overload is another big barrier. Small text, dense layouts and moving elements make it harder to succeed in tasks (Haesner et al., 2017; Castilla et al., 2018). Interfaces for this group therefore need large, readable text, minimal distractions, calm layouts and enough space between elements. Removing unnecessary content and pulling attention to the primary action reduces uncertainty. Instead use consistent layouts, descriptive labels, clear button texts and predictable interactions to help users maintain focus and confidence.(Jokisuu et al., 2011). Long or abstract instructions, especially those using inconsistent terminology or metaphorical labels, are harder to understand and should also be avoided.

Emotional impact and confidence

Emotional factors also play a strong role. Users with MCI often experience anxiety, fear of making mistakes, or a lack of confidence in digital environments (Castilla et al., 2018). This means interfaces should feel forgiving and supportive, with clear error messages, visible progress indicators, confirmation messages and simple recovery options.

Research Derived Accessibility Guidelines

Applying research findings to accessibility design

To support users with MCI, MillorIA should follow accessibility principles that address age-related functional changes and cognitive challenges. While the WCAG 2.1 standards provide a good foundation for accessibility, there are additional guidelines necessary to meet the specific needs of the target group. By combining WCAG compliance with additional considerations, the website can provide an accessible, supportive, and user-friendly experience for older adults and people with MCI, reducing cognitive load and increasing confidence while navigating digital tasks.

Visual Accessibility

Use high contrast between text and background; avoid low-contrast colors or glare from bright white backgrounds.
Ensure large, readable text and scalable fonts for easy reading.
Avoid small or complex visual elements; prioritize clear, simple layouts with ample spacing between items.
Limit visual distractions such as moving elements, pop-ups, or dense clusters of information.

These guidelines address age-related declines in contrast sensitivity, vision decline and visual processing speed. For users with MCI, unclear visual hierarchies or excessive visual stimuli increase uncertainty and scanning behaviour, which can lead to hesitation or errors. By prioritizing clarity and reducing visual noise, the interface supports faster recognition and lowers cognitive effort during interaction.

Hearing Accessibility

Provide captions and transcripts for all audio and video content.
Avoid relying solely on audio cues; include visual or text-based alternatives for important information.
Minimize background sounds or music that could interfere with comprehension.

Because hearing loss is common among older adults and is linked to increased cognitive load, relying solely on audio information can exclude users or force them to divide attention. Providing visual alternatives ensures that information remains accessible and reduces the risk of misunderstanding or frustration during use.

Motor Accessibility

Use large click or tap targets to accommodate loss of motor skills.
Avoid complex gestures or rapid sequences of clicks or dragging.
Ensure sufficient spacing between interactive elements to prevent accidental selections.

Age-related motor skill decline can make precise interactions difficult, increasing the likelihood of errors. For users with MCI, such errors may lead to confusion, anxiety or loss of confidence. These guidelines reduce the physical effort required to interact with the interface and support a more forgiving experience.

Cognitive Accessibility

Structure the interface with predictable navigation and a clear, simple flow.
Break complex processes into step-by-step tasks and provide progress indicators.
Use short, simple sentences and consistent terminology; avoid metaphors or abstract labels.
Reduce cognitive load by showing only relevant information and minimizing unnecessary options.
Avoid time-limited tasks or allow flexible timing.
Provide clear error messages, confirmations, and recovery options to reduce anxiety and support confidence.
Offer recognition-based authentication such as remembering user profiles to reduce memory demands.

These principles directly address the cognitive challenges associated with MCI, such as reduced working memory, slower processing speed, and difficulty with planning. By minimizing the need for recalling information and providing clear structure and feedback, the interface helps users maintain orientation and complete tasks without feeling overwhelmed or stressed

Technology Choices

Framework

For this project I will be using React in consultation with the team at Vertixlet. React is a JavaScript library used to build user interfaces for websites and web applications. It lets developers create reusable components: small, self-contained pieces of a page, like buttons, forms, or menus, that can be combined to build complex applications. React makes websites more interactive because it can update only the parts of the page that change, instead of reloading the whole page. I have chosen this platform for the simple reason that I am already familiar with it. It has a big online community and a lot of documentation accessible.

Another reason I prefer working with React is the ecosystem around it. There are so many tools and libraries that fit naturally with it, which makes it easier to shape a project the way I want. Because of this flexibility, I'm not tied to a strict structure or workflow and I can pick the tools that make the most sense for the project.

Database

For this project I looked into two databases, Firestone and Supabase, that are known to work well with React. Something that was important for this project was that the database was good with handling JSON formats since that would be the input and output for Isabel's API. Firestore has several advantages. It stores JSON-like documents natively, making it suited for dynamic AI-generated data. It integrates easily with React, offers built-in authentication with role-based security rules and provides real-time syncing. However, it has drawbacks: querying is limited compared to SQL, making complex relationships harder to manage.

Supabase, on the other hand, is a relational database with full SQL support while still allowing flexible JSON storage via JSONB. It includes built-in authentication and Supabase has built-in auth.users for authentication.

You can extend this with a profiles table to manage roles, which is ideal for managing admin and users access. Supabase also makes it easier to manage structured relationships, such as linking exercises to practicers. Its drawbacks are that it is not real-time by default, although subscriptions are available, and it requires more setup and database knowledge than Firestore.

Role-Based Account Structure

There are two types of users on the platform: Caregivers or family members (supporters) and individuals with MCI (participants). Having two types of users requires a clear separation of roles within the platform. And this separation starts when a user registers. From the start, I considered several options for registration, weighing the pros and cons of each.

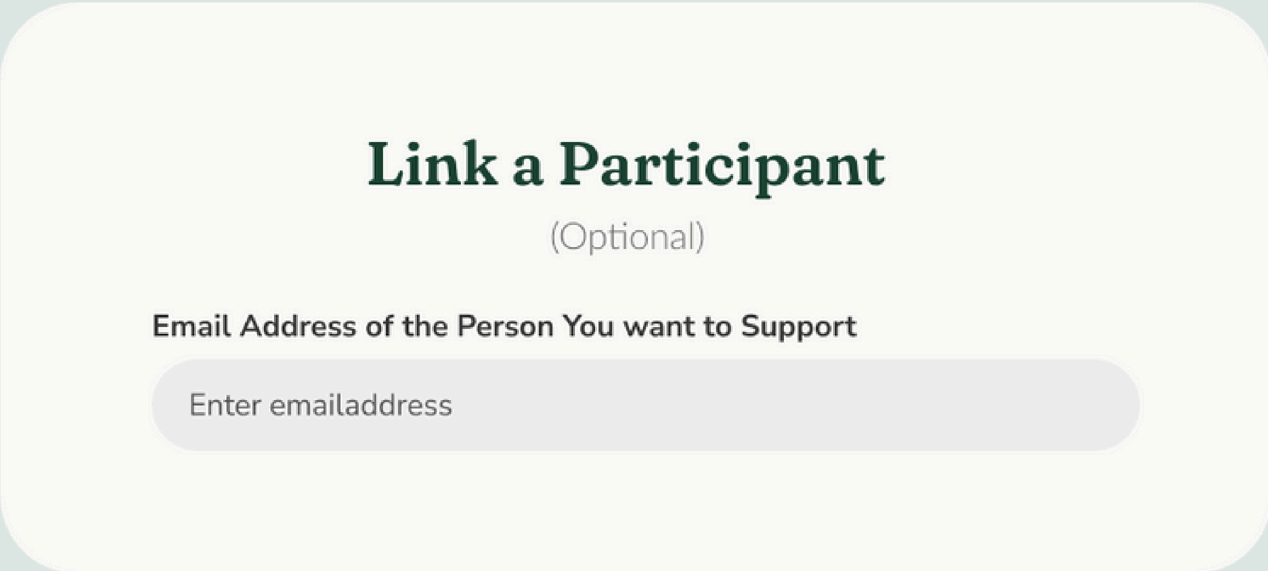
One option was for supporters and participants to share a single account. This approach would allow them to complete exercises together and would be relatively easy to implement. However, it would limit independent use of the platform, as participants would need access to the same login credentials. Since passwords are sensitive information, it cannot be assumed that everyone involved would be comfortable sharing them. This is particularly true in a professional caregiver–patient setting, where shared credentials could raise privacy and security concerns.
A variation on this was having one account but separate profiles for supporters and participants. An example of this being done is Netflix, or almost every other shared streaming platform.

Example of a shared account with separate user profiles, as used by Netflix. Each profile represents a different user within the same account, allowing personalized access while sharing login credentials.

This would make it easy for supporters to help participants navigate the platform, though it would be slightly more complex to implement. This would give us the benefit of being able to shield certain functions for the participant to keep complexity down. However, sensitive information would still have to be shared.

The approach I went for is separate accounts. This would keep sensitive information private and allow supporters to have one account with multiple participants linked to it. With separate accounts participants could complete exercises independently while supporters were able to manage their accounts. The challenge with this approach was to make it as simple as possible for supporters and participants to link their accounts.

I looked at different ways to link participants and supporters. The first was code-based linking, where each supporter has a unique code participants enter during registration. This is fully self-service but can be less secure if codes are shared. Also it requires the participant to write down or remember a code and navigate to the website to register. This is something that could be a painful process for someone who can handle a smaller cognitive load and was advised against in the research findings. The option I chose to implement was invite-based linking: everyone registers freely, choosing their role and supporters can later invite existing or new participants by email. This keeps registration open but requires an extra invite step and still some independence from the participant who has to register themselves before or after being invited. Also instead of linking them after registration I gave the supporter the option to invite a participant during his registration process.

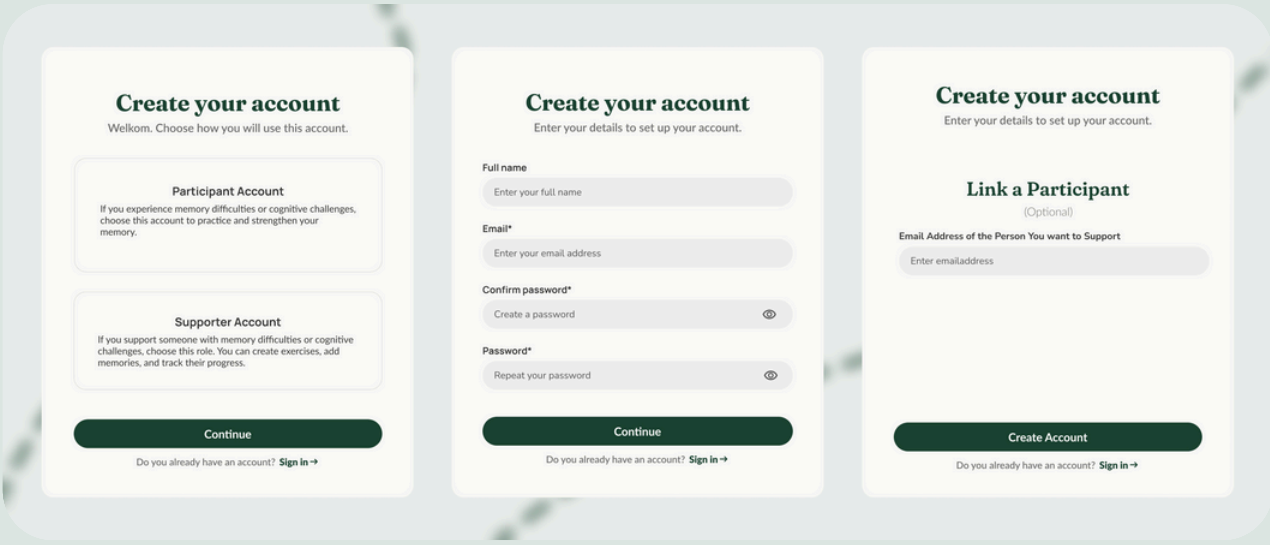


Input field at the end of the register flow, allowing supporters to invite a participant

When a participant is invited, a unique token is generated and sent via email. This email directs the recipient to the login or registration page, with the token included in the URL. When logged in or registered successfully, the database uses this token to automatically link the participant to the corresponding supporter account.

The same technique is used when a supporter is linked to a participant, allowing the supporter to gain access to the participant's account. In both cases, the token ensures a secure connection between accounts without requiring shared login credentials.

Although this was the most suitable option for linking participants and supporters, it introduced additional UI complexity. The registration interface in particular required a considerable amount of information, including selecting the correct role, entering an email address, password, and name, and in the case of a supporter, the option to invite a new participant. For that reason I chose to divide the register form into a multistep form to reduce overload. This choice was in line with research showing that breaking down complex tasks reduces cognitive load and minimises errors among participants with memory and attention issues.



Three step register form

Design Setup

User personas

User personas are fictional profiles that represent key features of the target group. I created the user personas to clarify who I was designing for and to better understand their needs. Defining the personas early helped ensure that design and development decisions aligned with real user goals. You will find the user personas on the next pages.

PARTICIPANT PERSONA

A QUICK PROFILE OF MY PRIMARY TARGET GROUP

PROFILE

Name

Maria García

Age

72

Gender

Female

Location

Spain

Occupation

Retired primary school teacher

Background

Living situation: Lives independently, but her daughter visits a few times a week to help with some household tasks.

Technology use: Uses a tablet for photos, games and video calls; sometimes gets confused with error messages or navigating the interface.

PERSONAL

Health and cognitive profile

- Mild Cognitive Impairment (MCI) diagnosed 7 months ago
- Sometimes forgets names, words, passwords, or how to navigate websites she doesn't use often.
- Vision and fine motor skills have slightly declined because of age

Needs

- Clear, consistent navigation
- Large buttons, readable text, and familiar icons
- Encouraging feedback to stay motivated

"Sometimes I just need a little reminder of how to do things. Once I see it, I remember."

GOALS

Primary goals

- Stay mentally active and maintain independence
- Keep her mind sharp through memory exercises

CHALLENGES

Frustrations / Challenges

- Gets frustrated when instructions are unclear or text is too small
- Feels anxious when she "gets lost" online or can't find her way back to the home page
- Dislikes being treated as incapable

SUPPORTER PERSONA

A QUICK PROFILE OF MY SECONDARY TARGET GROUP

PROFILE

Name

Laura Martínez

Age
42

Gender
Female

Location
Spain

Occupation
Judge

Background

Living situation: Lives 20 minutes away from her mother, visits several times per week

Technology use: Is very competent with all sorts of technologies and uses it daily for her job and personal life.

PERSONAL

Heallth and cogniitiive profiille

- Has no health issues

Needs

- Easy-to-use caregiver dashboard
- Option to test exercises before assigning them
- Progress insights to track her mother's engagement

"If it's easy for me to set up, I'll keep using it. If it helps Mom feel connected, it's worth it."

G OA LS

Primary goals

- Help her mother stay engaged and independent
- Help her with the exercises where needed.
- Upload family photos and memories to support memory.

CHALLENGES

Frustrations / Challenges

- Limited time to help set up or troubleshoot technology.
- Sometimes unsure which exercises are suitable for her mother's abilities.

Participant User Stories

Participant can register
As a participant, I want to register easily and be able to create an account linked to a supporter.

Participant can log in
As a participant, I want to be able to login easily so I can start my session without confusion or frustration.

Participant can log out
As a participant, I want to log out when I'm done so that my information remains private and secure.

Participant can view memories
As a participant, I want to browse through my personal memory collections with familiar photos, sounds, and stories, so I can recall positive moments and stay emotionally connected.

Participant can do exercises
As a participant, I want to complete personalized exercises based on my memories, so I can train my memory in a way that feels familiar and enjoyable.

Participant gets validated
As a participant, I want to receive clear validation feedback when the information I enter is incorrect or incomplete.

Supporter User Stories

Supporter can register
As a supporter I want to be able to register into a separate but linked account to the participant.

Supporter can log in
As a supporter, I want to log in securely to the platform so that I can access the memory collections and exercises linked to the person I'm supporting.

Supporter can log out
As a supporter, I want to log out easily so that personal information and uploaded memories remain private and secure after a session.

Supporter can upload memories
As a supporter, I want to upload personal memories in the form of photos, text or audio clips that are meaningful to the participant, so the AI can generate personalized exercises based on those memories.

Supporter can upload audio, images, and text
As a supporter, I want to be able to add different types of media (audio, images, and text) when creating a memory.

Supporter can edit memories
As a supporter, I want to edit existing memories so that I can correct mistakes or add new information.

Supporter can delete memories
As a supporter, I want to delete memories that are no longer relevant for the participant, ensuring that the memory collection stays up to date and positive.

Supporter can view memories
As a supporter, I want to view all the uploaded memories in a clear and organized layout, so I can review what has already been added and see what might still be missing.

Supporter can do the exercises
As a supporter, I want to be able to access and try out the exercises myself, so I can understand how they work and ensure they are appropriate and easy to use before introducing them to the participant

Supporter gets validated
As a supporter, I want to receive clear validation feedback when the information I enter is incorrect or incomplete.

Supporter can add new participants and additional supporters
As a supporter, I want to be able to link participants to my account and add other supporters to a participant's environment, so they can also help manage the participant's account.

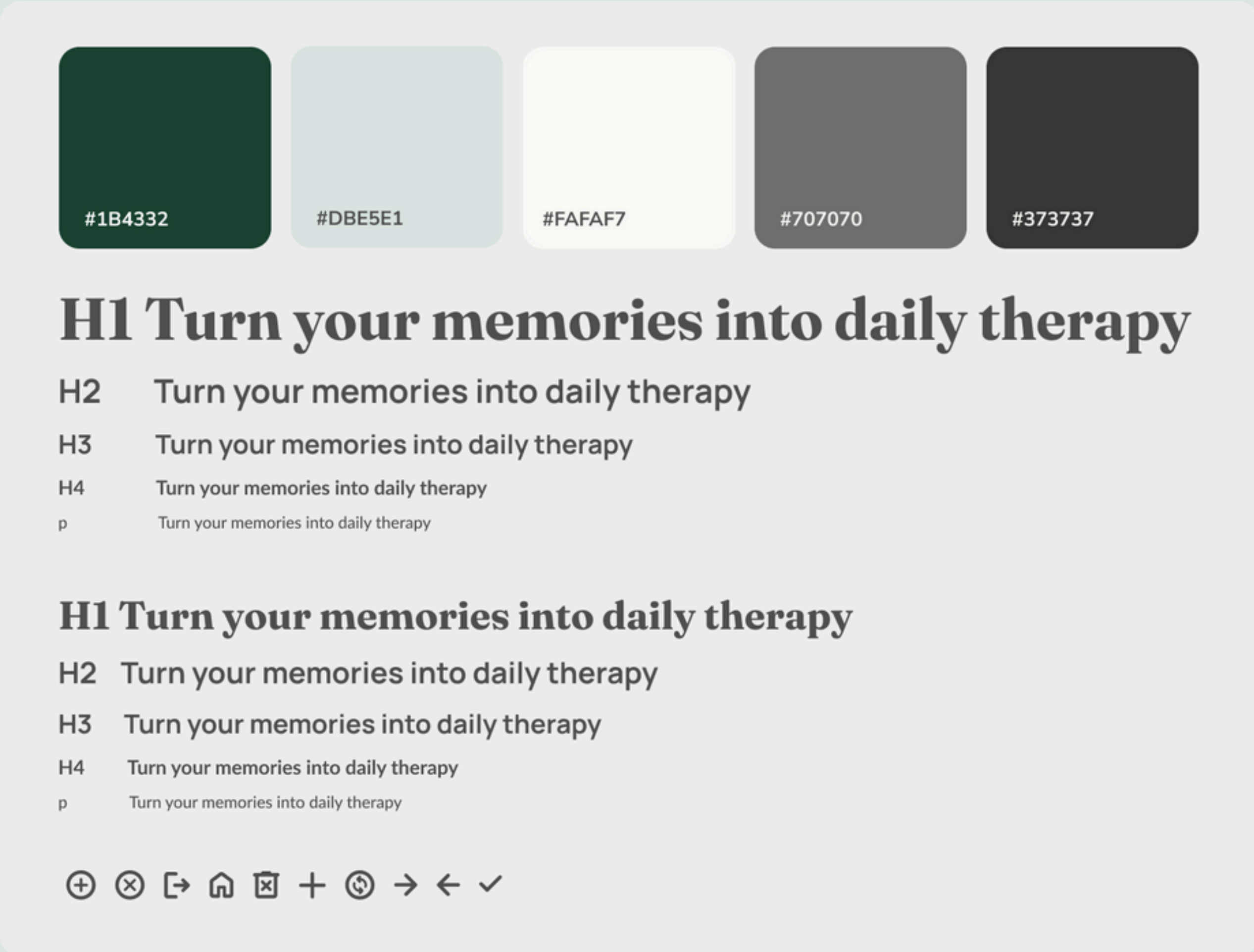
Supporter can adjust the dashboard of the participant
As a supporter, I want to be able to turn on or off certain functions for the participant to avoid unnecessary complexity.

User Values

The research into the target group revealed several core values. For people with MCI, clarity, predictability and security are important in reducing anxiety and uncertainty. For supporters, control and a clear overview are essential for handling personal information and memories. These values formed the basis for further design choices.

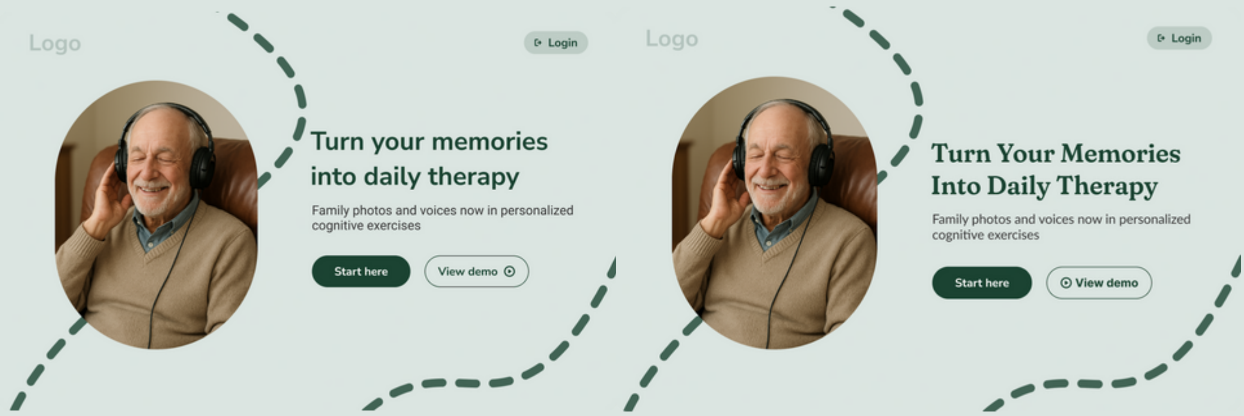
Design Process and Visual Design Decisions

I started with the design system. I chose a single primary color to keep the overall look calm and consistent and a light green as secondary for details and backgrounds. The color palette consists of an off-white with a subtle yellow undertone and a muted black. I made this choice deliberately, as my research showed that high-contrast combinations, like pure black on bright white, can cause eye strain for older adults.



For the icons, I used Google Fonts icons. They are straightforward, simple fonts and are easy to customize, so I adjusted them to be slightly thicker and more rounded to match the soft, rounded aesthetic of the interface.

For the main header font I chose Fraunces. Fraunces is a serif font with a soft, friendly, slightly retro look. It has noticeable curves and contrast in the strokes, which makes it feel warm and expressive. It works well for headings because it adds personality and a bit of elegance without looking old-fashioned. It gave the site the bit of elegance and warmth it was missing. Before this, I was using Nonito Sans as the title font. But that font lacked the elegant feel I wanted. It looked calm but also plain and did not match the warm and refined environment I had in mind.



Left the design with the font Nunito Sans and right the final pick with the font Fraunces

For the body text and titles I picked Lato. Lato is a clean, modern sans-serif font. It feels neutral, smooth, and is easy to read. The letters have gentle curves that make the font feel friendly instead of strict. Because it’s so readable, it works well for body text and long paragraphs.
When designing the website I imagined the atmosphere of a warm and elegant retirement home. Think of dark brown wooden furniture with a polished, slightly glossy finish. Soft, warm lighting that creates a peaceful glow in the room. Comfortable armchairs. Classic framed pictures on the walls. Large patio doors that look out onto a beautiful English garden. The kind of place that feels peaceful, welcoming and cared for.

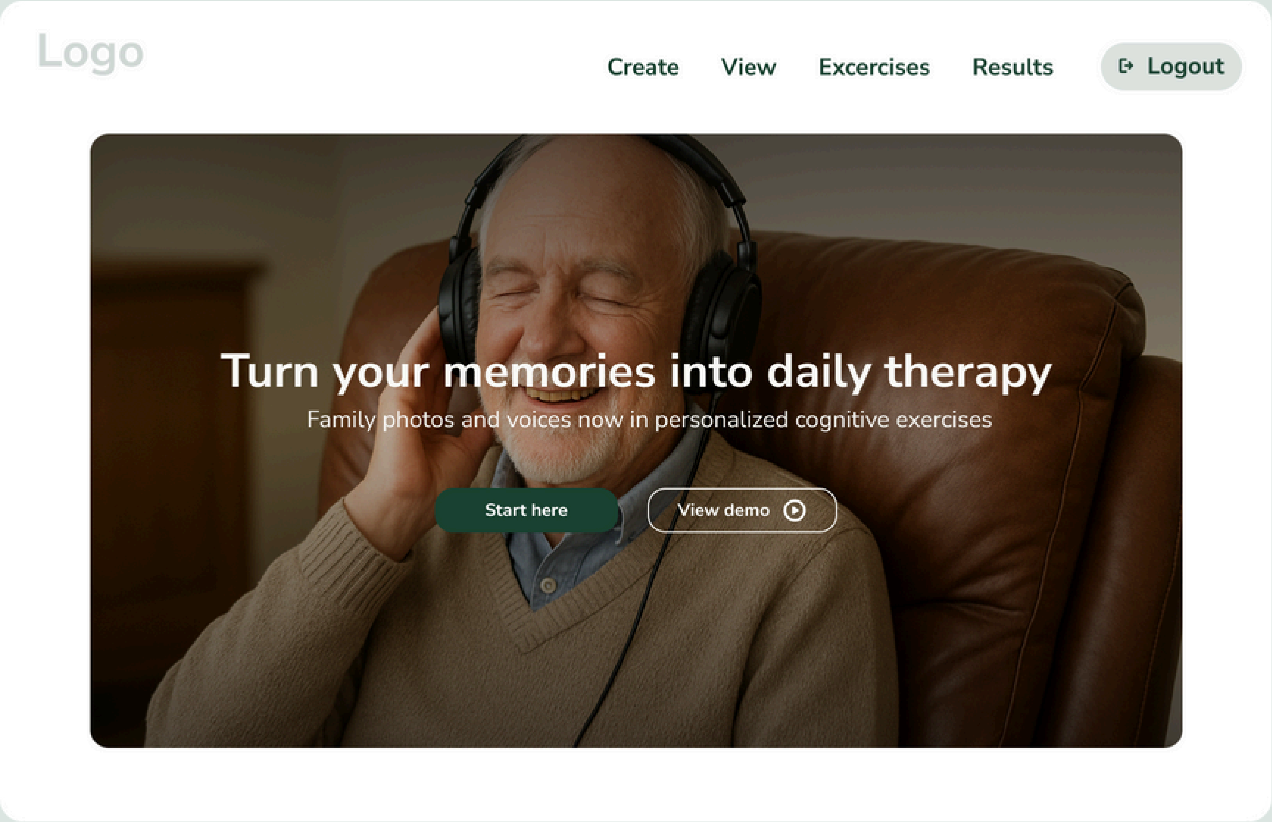
My inspiration for the design

The Homepage

The designing of the homepagebegan with low-fidelity mockups. I used these to outline the general layout and positioning of the main components. I didn't create low fidelity sketches for every element, because I find them less useful for components that contain a lot of detail, such as forms or login screens. For those, I prefer to design directly in higher fidelity, since it helps me to better see whether something will actually work visually and functionally.

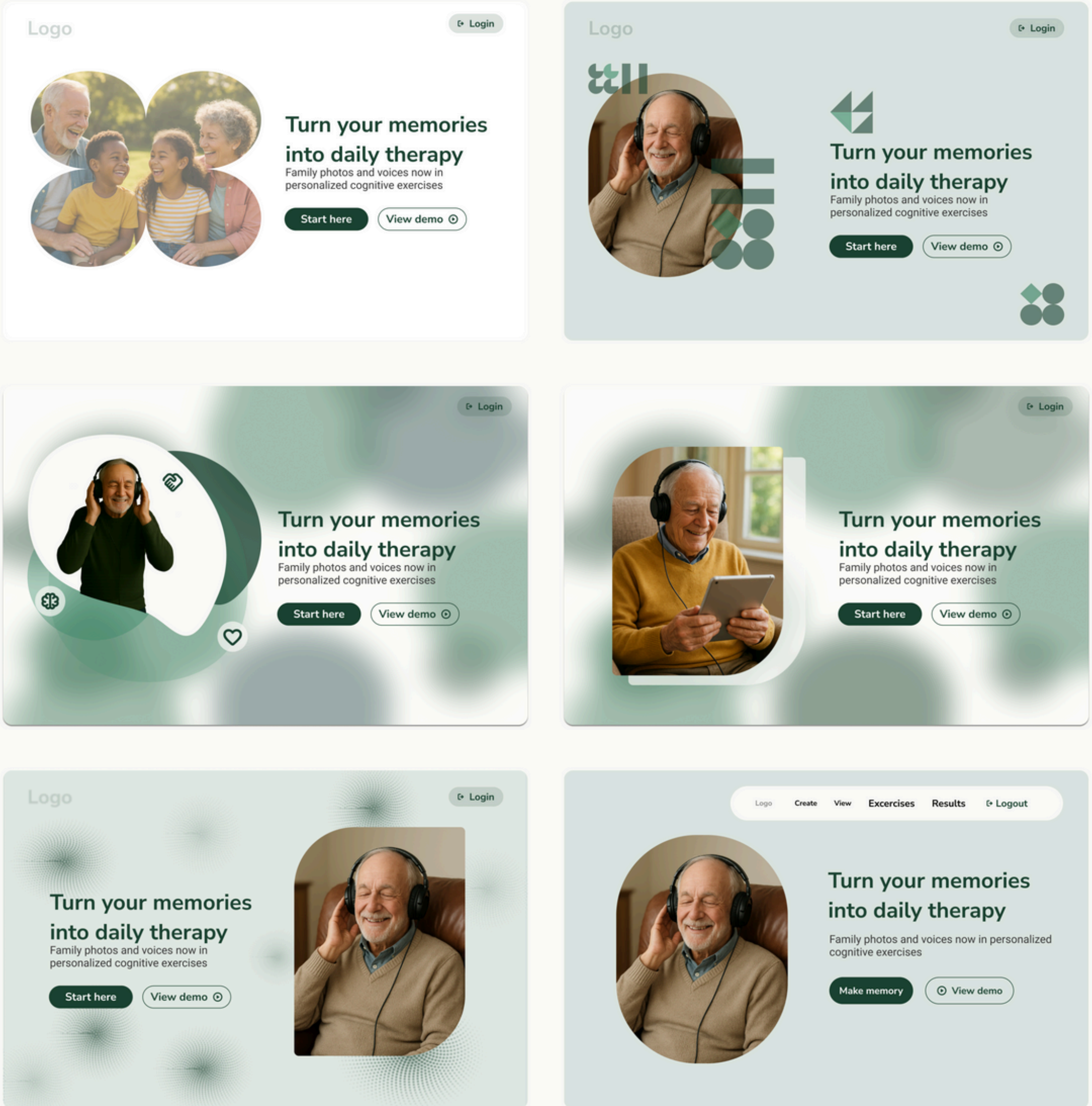
For the low fidelity sketches I made two options for the homepage. This was option one:

And I turned it into this high fidelity mockup.



This mockup was visually unappealing and placed text over images, which research showed can be harder to read for some users.
I made a second low fidelity mockup where the text and image were separated.

This would eventually turn into the final design. But not after a lot of other trials:

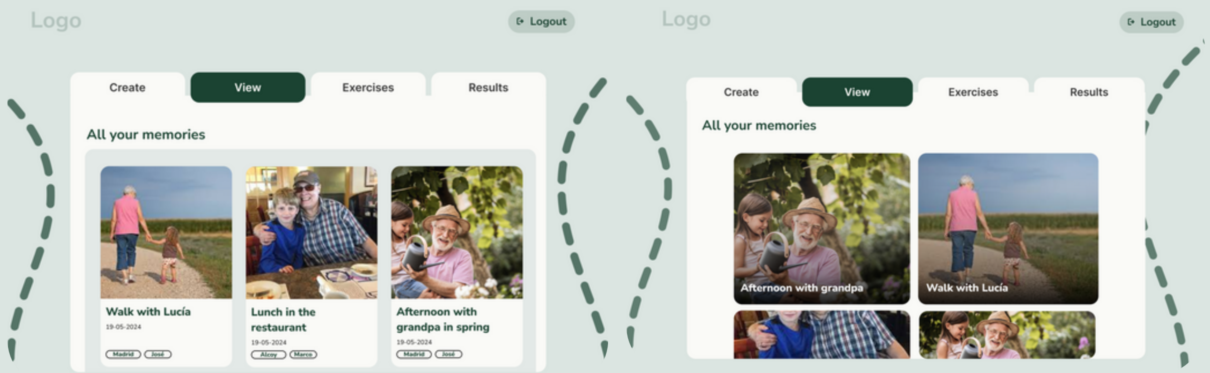


The Memories view

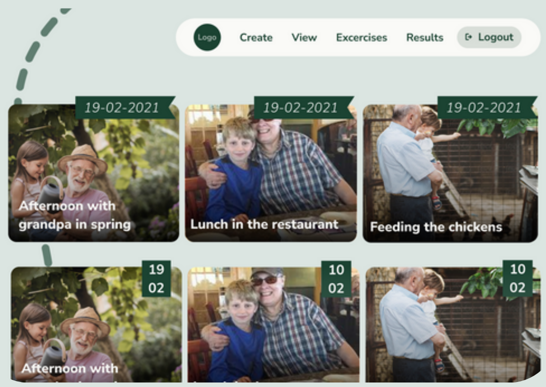
The view went through many iterations. My goal was to create a comforting, nostalgic feeling, like a trip down memory lane. I started with timeline-based mockups to show the passing of time in memories. However, this layout only fitted two memories on the screen at once. And to view all the memories, if a participant had more than a couple, would require way too much scrolling. I had some exciting ideas as well using scroll-based animations to reveal different years and moments, however in the end I decided against this because in my research I concluded that large animations could distract participants and create confusion without adding real value. I then shifted to a more traditional grid layout.

Mockups with timeslines

Early versions also included a folder-style tab container instead of a navigation bar, but this limited how much content could be displayed on the screen. Everything had to fit inside a small card, which made the layout feel constrained and messy.



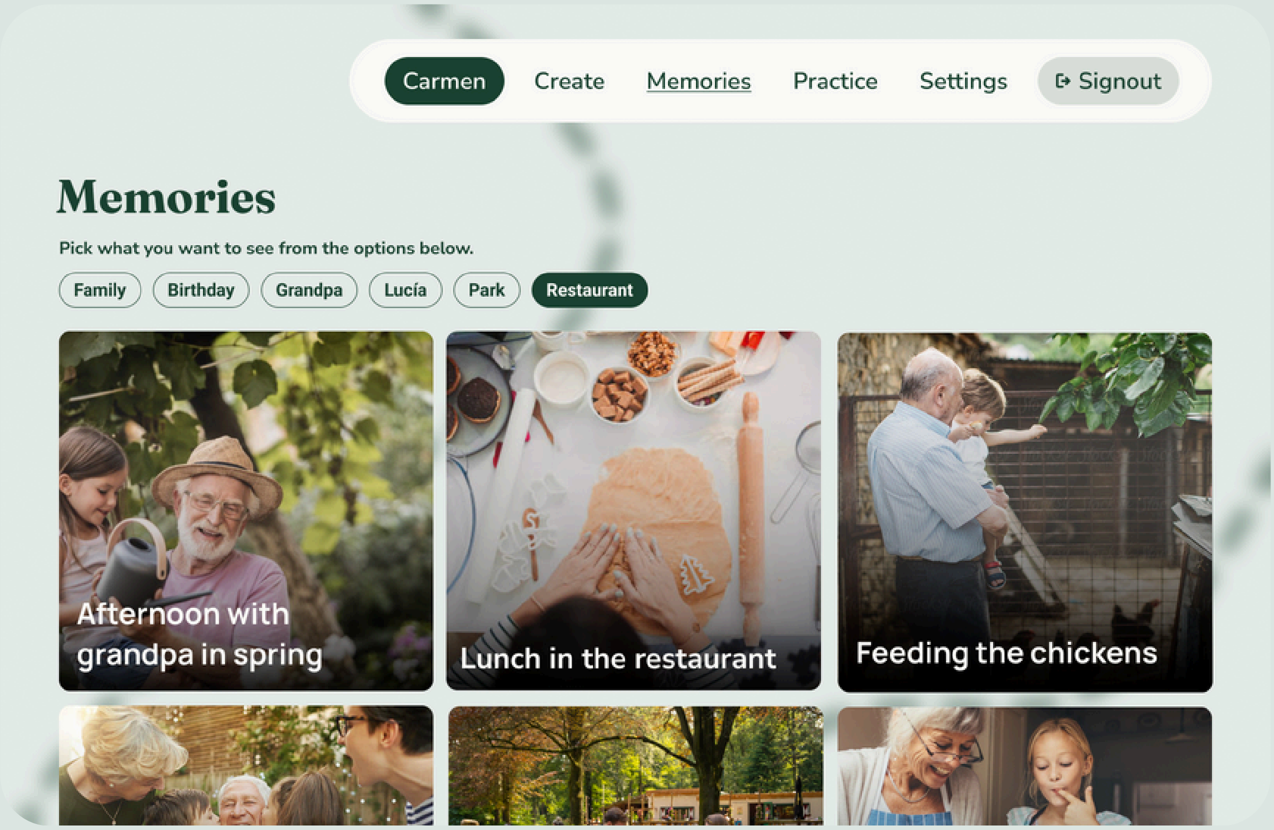
Mockups of the view with a folder layout



Removing the folder layout allowed the design to use the full screen, bringing it much closer to the final version. I also experimented with showing dates, but during a conversation with Isabel it became clear that the focus should be on the memories themselves rather than exact timestamps which were not an important part of the memory.

One the left: A mockup showing dates

Removing the dates resulted in the design below: simple, clean, and focused on the content. The images are large so that participants with poor eyesight can easily view and recognize memories without needing to click on them.



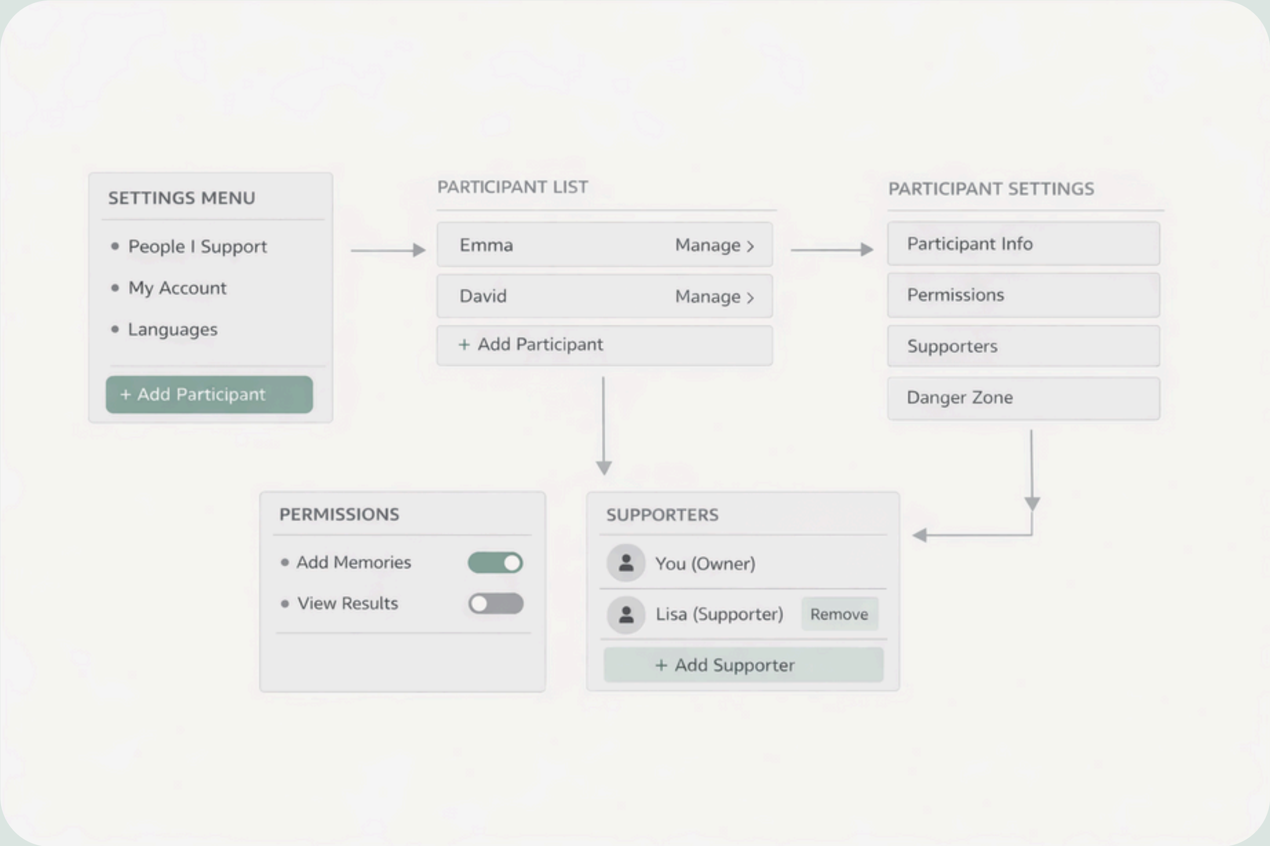
After testing this design I found that it still needed a couple changes. These changes were to make the design more accessible for the user group with mild cognitive impairments. In the chapter Usability Testing I will elaborate on these changes.

The Settings Page

The settings page was the most challenging part to design. It had many layers, buttons and links to take into account, and keeping everything simple and easy to understand was difficult.

The system also had to work for two types of users: participants and supporters. Every participant needed their own settings page, but supporters also needed extra functionality to manage the people they support. This included switching between multiple participants, editing their general information, managing their permissions, unlinking a participant from their account and adding additional supporters. Designing all of this in a way that remained clear required many iterations.

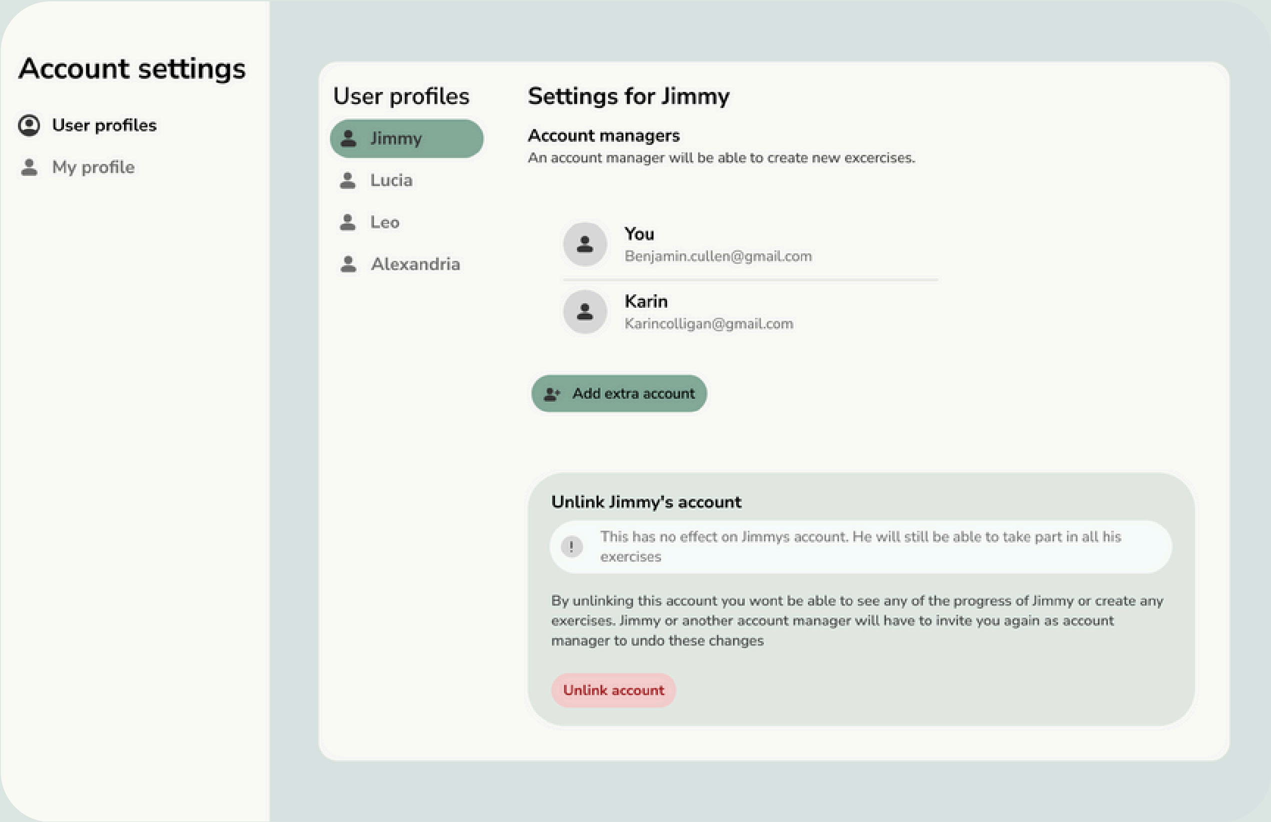
Creating the settings page strongly confirmed the decision to create separate accounts. By separating participant and supporter accounts, it became possible to present a simplified interface to participants, while reserving more complex options and controls for supporters.



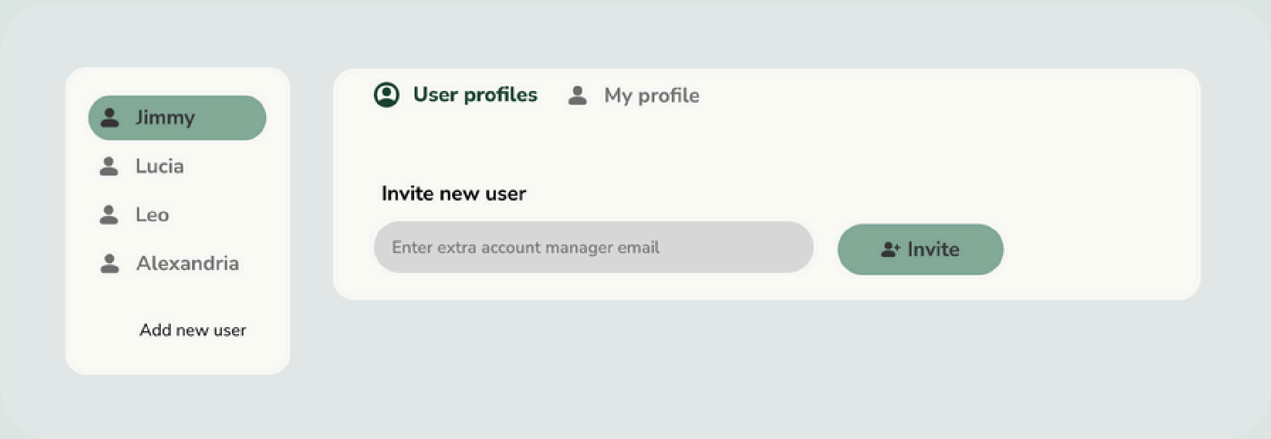
The idea was to create a dashboard with two layers for supporters. One layer is for managing their own account, such as editing their name, email, password, or deleting their account. The second layer is for managing the participants linked to their account. Participants themselves only see their own account, which they can edit, and do not have access to the supporter layer.

One of the main challenges was keeping the interface clean and not visually overwhelming, while still making important actions, such as adding a participant, easy to find. Switching between participants also needed to feel natural and intuitive, without making the interface more complex. Balancing all of this while keeping the design looking calm, professional and easy to use required a lot of ugly designs before landing on the end result.

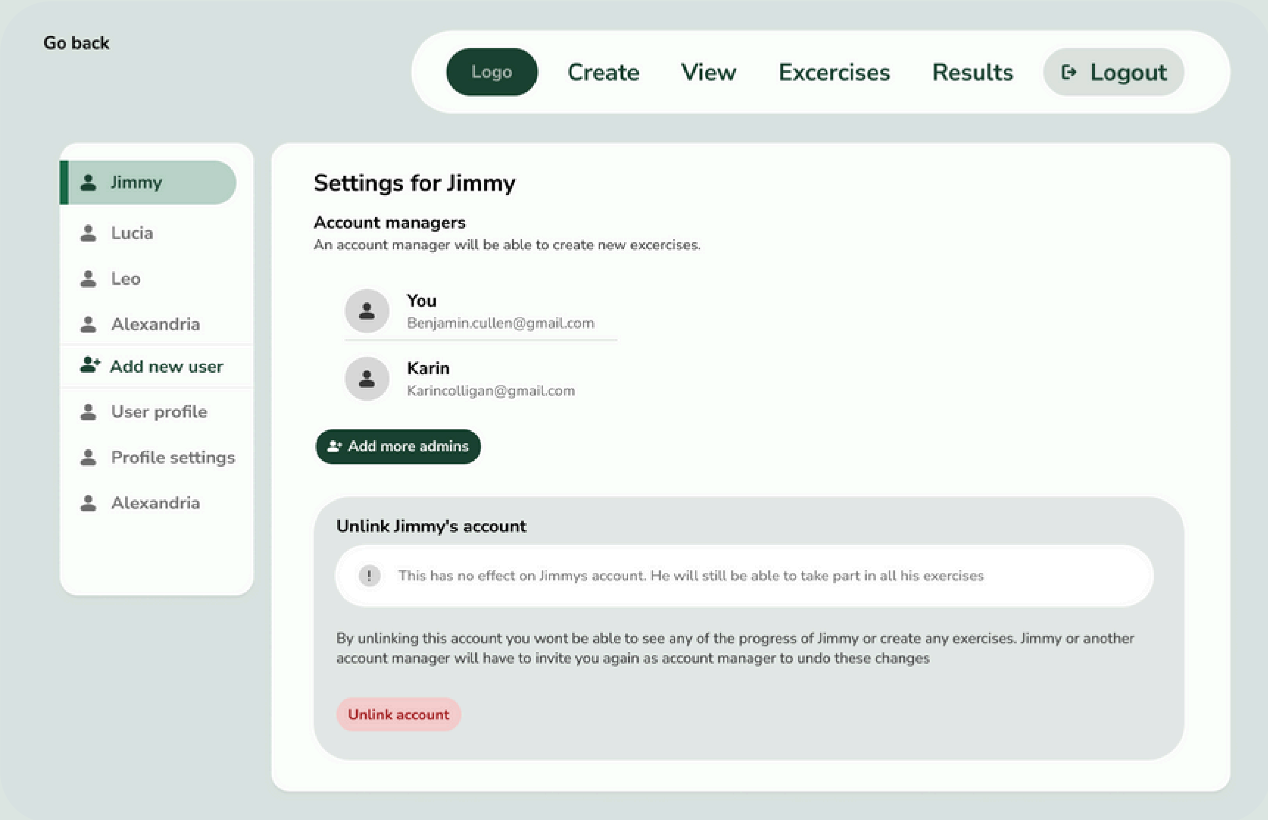
Feedback from usability testing helped me in iterating the design. It highlighted which settings were overlooked or misunderstood, such as adding supporters or switching between participants. These insights impacted the hierarchy of the settings and made me change the wording in certain labels in later iterations.



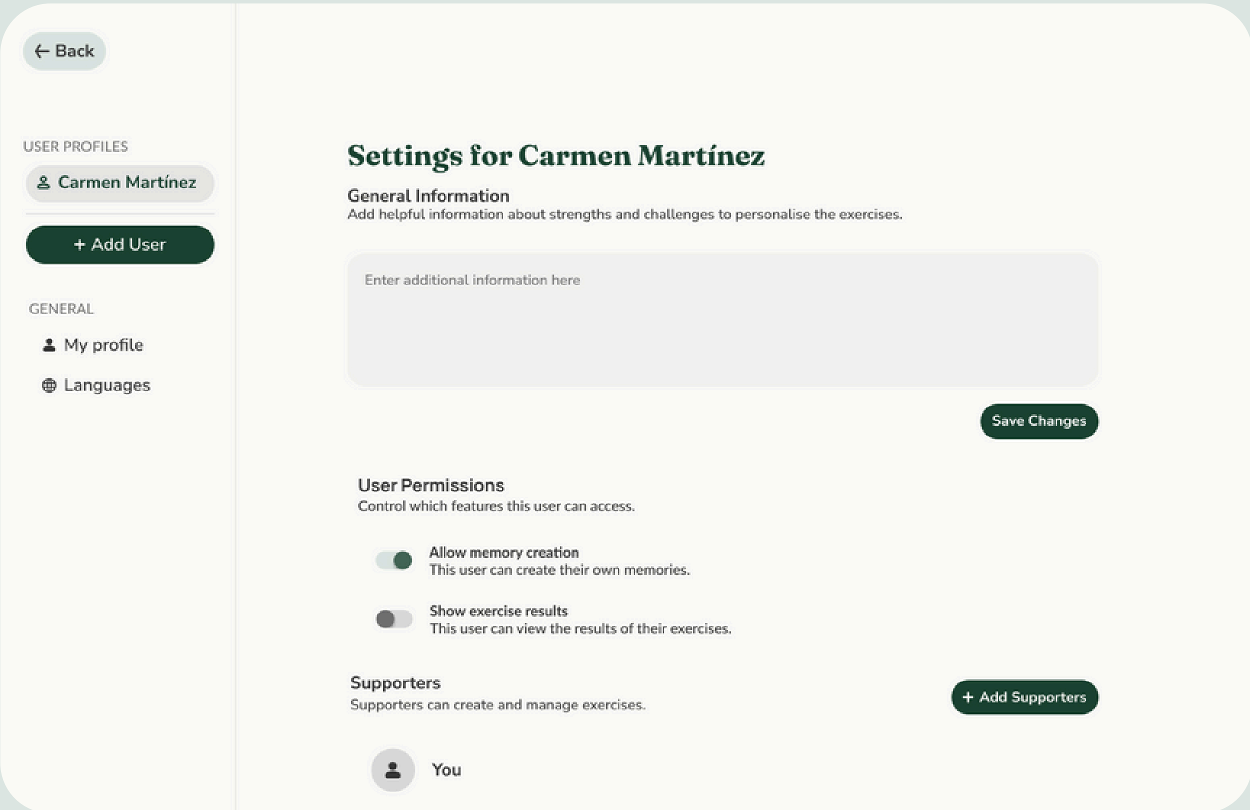
One of the first mockups with stand-alone sidebar

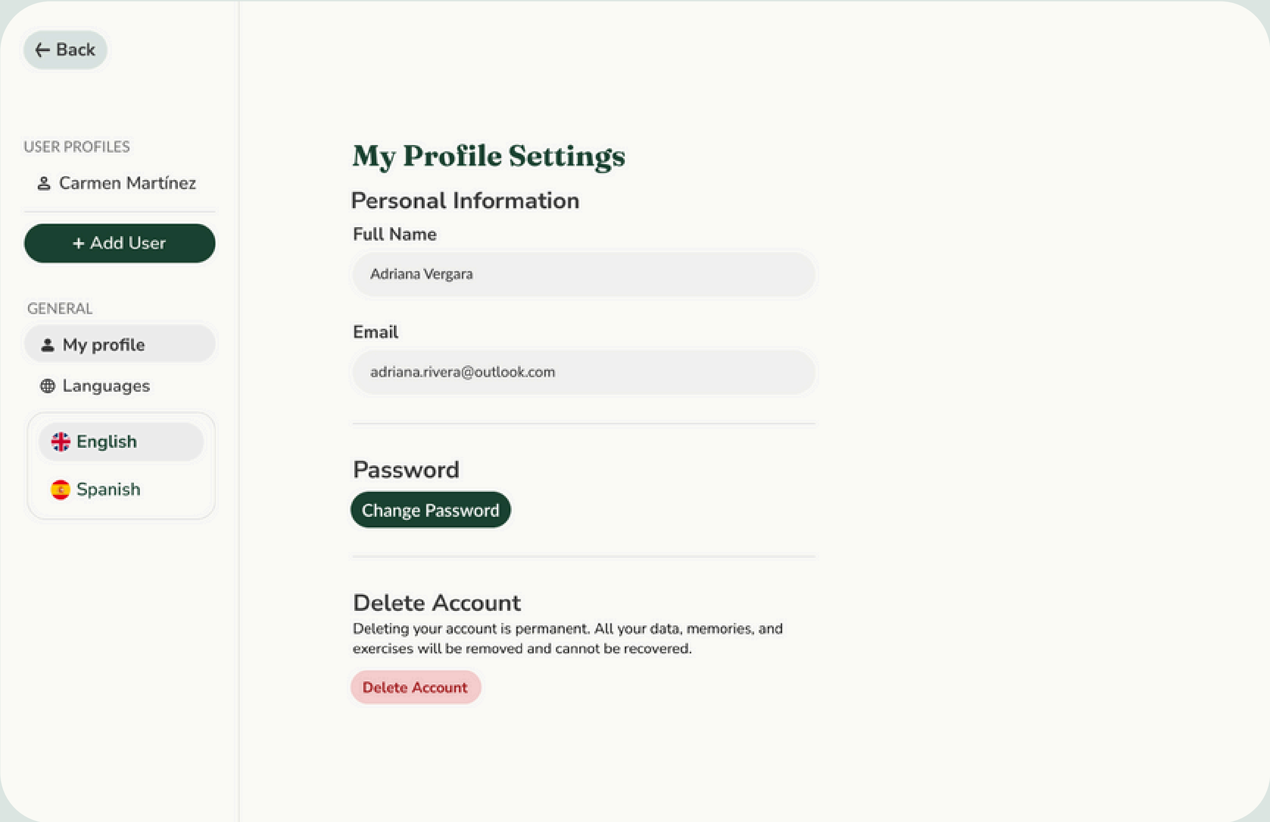


Mockup where the user profiles and my profile are at the top instead of in a sidebar



Mockup with all the information in the sidebar





Final designs of the settingspage. The user can switch between the participants view and their own view of the settingspage.

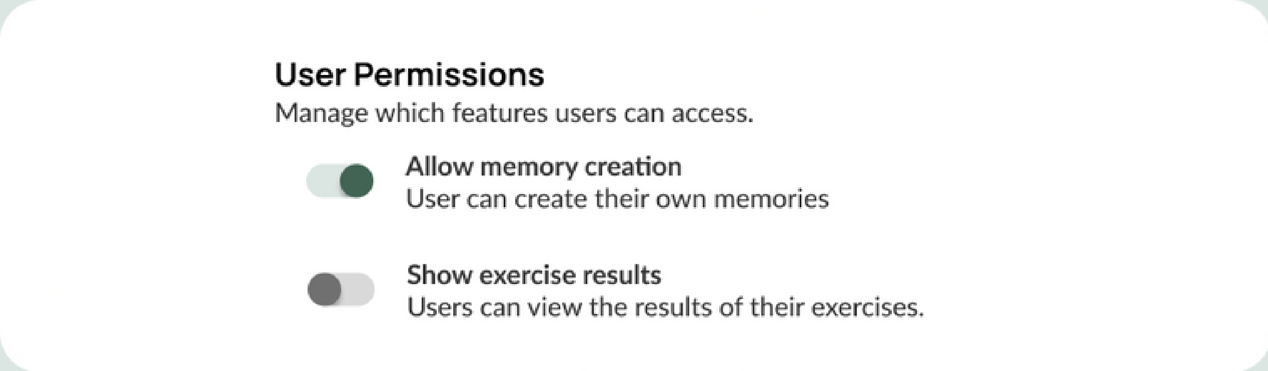
Expert Feedback: Neurologists Perspective

On 19 November, Jordi, Isabel, and I met with José Manuel Moltó Jordá, a neurologist, at the hospital of Alcoy to discuss our prototype. The prototype was at this point a website with a functioning authentication and CRUD (Create, Read, Update and Delete).

The neurologist has extensive experience with patients with dementia and gave practical feedback based on his experience. He explained that dementia patients retain different abilities, so the focus should be on exercises that strengthen preserved skills rather than trying to improve weaker areas. For patients with dementia or correctly diagnosed MCI, the goal should be maintaining abilities and not recovering lost ones.

Regarding expectations, he emphasized that cognitive exercises can help keep patients engaged and reduce behavioral problems, but they should not be presented as therapy. The main benefit is maintaining abilities and improving quality of life.

We asked whether parts of the website should be limited for patients to avoid complexity. He said it depends on the stage of the disease. MCI patients usually like having choices, as they are not as far progressed in the disease, while more advanced patients may need a simplified interface. A practical solution would be for supporters to control what the patient can access.



Design based on the feedback to have supporters manage participants' views.

At the end of the interview the neurologist explained that it is difficult to determine whether the platform will be truly beneficial for participants, because evaluating effectiveness in people with MCI is complex. A reliable assessment would require tracking patient performance over a long period of time, asking simple well-being questions and possibly comparing results with similar patients who do not use the platform. On top of that, because there is no standard rate of cognitive decline, individual differences would always be expected and as such short-term improvement or stability cannot be measured in a trustworthy way.
This feedback helped me redefine the scope of my work. Within the scope of this project, there is neither the time nor the resources to conduct long-term studies, work with control groups, or gather medical data. At that point in the project, we were already deeply involved in development without knowing whether the platform would ultimately have a measurable clinical impact. Hearing this helped me rethink the scope of my work and set clearer boundaries. My role is not to prove or test the medical effectiveness of the platform, but to design and build an interface that makes such a platform usable and accessible. The key questions are whether people with MCI can understand the interface, complete tasks, don’t feel confused or stressed and can navigate the platform without constant help. These factors determine whether the platform can realistically be used in daily life, which is a necessary condition for any potential benefit to exist at all.

Usability Testing

For the user interface test I am testing on both supporters and participants. Supporters will create a supporter account to help their imaginary grandma with MCI navigate the platform. For the first round of testing I had created a clickable prototype in Figma that allowed enough interaction to test whether the design was intuitive enough and what still needed improving. After getting a decent amount of feedback from the first group I updated the prototype and tested the newer version on the second group. The third group even tested with the actual platform. In total I have tested on about 15 supporters.

The participants were a significantly smaller group. At the very last minute, we received confirmation from the neurologist regarding available testers for our participant target group. This resulted in a group of four testers with varying ages and stages of cognitive decline: one couple aged 80+ with mild symptoms of MCI, one man aged 70+ with early-stage Alzheimer’s disease, and one woman aged 65+ diagnosed with Alzheimer’s disease but with very mild symptoms.

Although not all participants fell strictly within the MCI target group, their symptoms closely overlap with those associated with MCI. Recruiting a test group consisting exclusively of people with MCI would be extremely challenging, as many individuals in this group are not yet under hospital care and may only be in contact with a general practitioner, or not receiving medical support at all.

All participants attended the hospital together with a family member or spouse who acted as a supporter which we could test as well. As all testers were Spanish or Valencian speakers, the interviews were conducted by Isabel, while I focused on observation and note-taking.

Test Goals

- Can testers easily find what they are looking for?
- Is the meaning of the different user roles clear enough?
- Do testers understand the navigation?
- Where do testers hesitate or get confused?

Thinking Out Loud Method

During the test, participants were asked to say out loud what they were thinking while using the interface. This included what they expected to happen, what they were looking for, or when something felt confusing. If they went quiet, they were gently asked to keep talking.

Tasks

Register

- If this is your first time on the website and you do not have an account yet, what would be your first step?
- Create an account.

Navigation

- You manage multiple participants. How do you switch to another participant?

Create, edit and delete

- You want to create a memory. What would you do?
- You made a mistake and want to edit or delete the memory.

Practice

- You want to start an exercise.

Settings

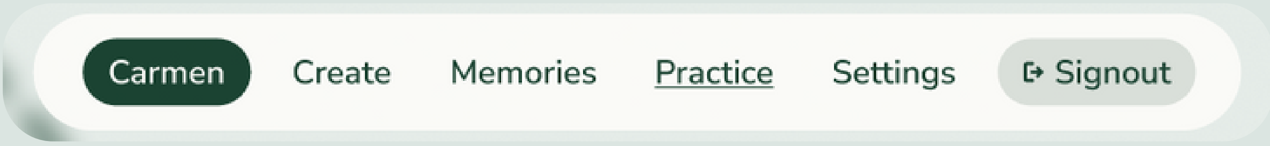
- You notice you made a typo in your name during registration. Try to change it.
- You want to add another participant to manage.
- You want to change the language of the platform.
- You want to change your password.
- You want to add another supporter.

Supporter Testing

Switching between participants

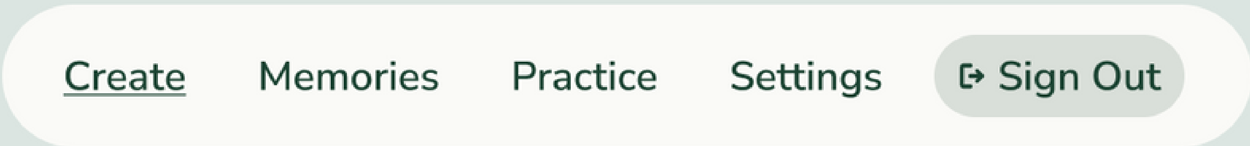
Issue

Several testers found switching between participants via the toggle button in the navigation unclear. Some did not recognise the button as interactive, while others were unsure which account was currently active and interpreted the displayed name as their own account rather than the participant they were managing. Almost all testers attempted to change participants via the settings page instead. Which was also a valid option.
This behaviour indicates difficulty with the understanding of “who is active,” which would be especially problematic when supporters manage multiple participants. Because selecting the active participant directly affects whose memories are viewed and created, clarity and certainty are very important for this feature. Based on this insight I changed the design.



Solution

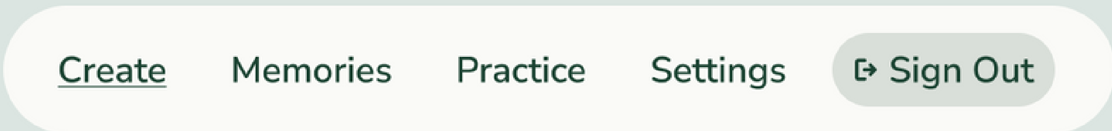
Remove the button entirely. This change was implemented in the next iteration of the prototype and validated. Testers no longer expressed confusion about account switching in the navigation.



Create in navigation

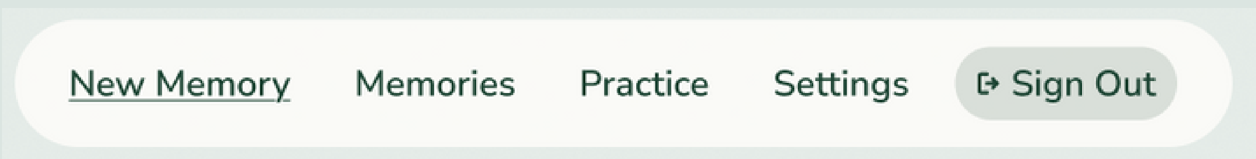
Issue

One tester did not immediately understand the purpose of the Create link in the navigation. Some testers would try to use the link to add participants or supporters as well. The label *Create* was considered too vague and open to interpretation.



Solution

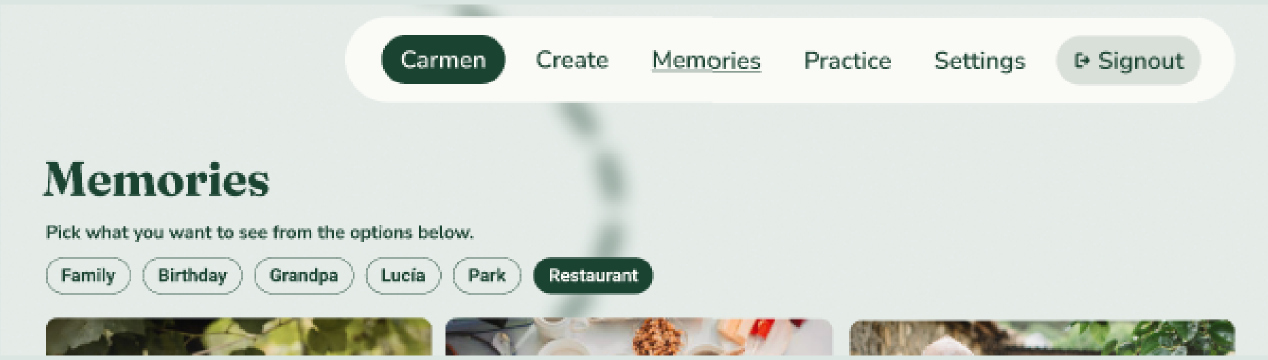
Rename *Create* to a more descriptive two word label such as **New Memory**. This makes the intended action clearer.



Memory page

Issue
Some testers did not realise they were on the memory page after creating a memory. This was likely caused by the prototype not providing clear feedback in the navigation on the current location and the absence of a clear page title, making it difficult for testers to orient themselves within the interface. One of the research findings also pressed on the importance of clear navigation and orientation so it was important that this issue was fixed.

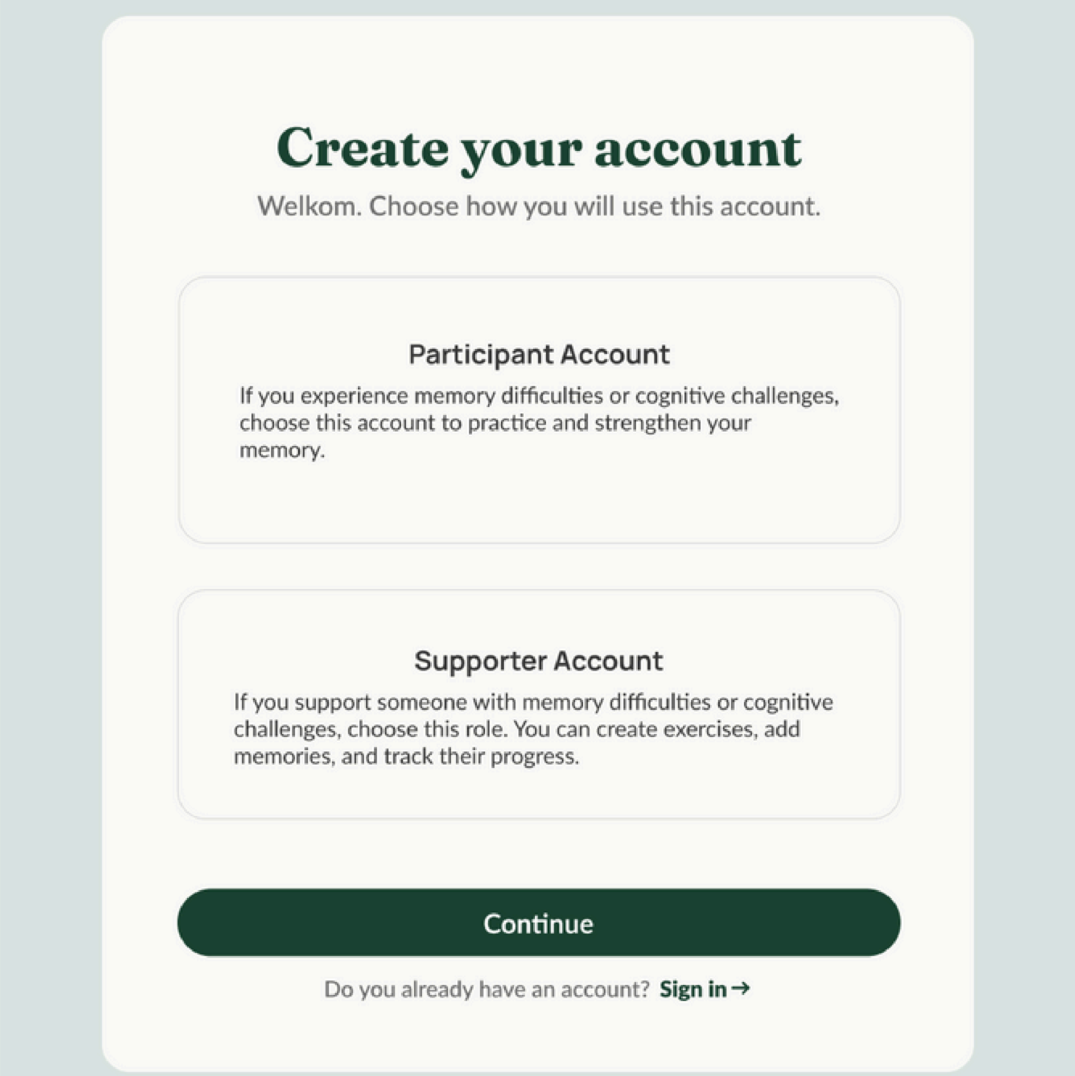
Solution
Add a visible page title to every page and clearly indicate the active page in the navigation. In the second prototype, the active navigation item was underlined and a page title was added. During later test sessions, this issue no longer occurred, indicating that these changes improved the testers orientation.



Unclear role naming

Issue
At this point of the project the participants were still called users and several testers misunderstood the explanation of the different user roles. As a result, they registered themselves as a *user*, reasoning that they were a user as well. While this reasoning is understandable, it led to incorrect account setup. This confusion showed that these role names were too abstract.The label *user* was too generic and did not clearly communicate the intended role within the platform. This confirmed the research findings that recognizable and concrete terminology are very important, which led to renaming roles and adjusting the explanations.

Solution
Change the role naming to more descriptive, non medical terms that better reflect the purpose of each role. Medical terminology such as *patient* or *caregiver* was intentionally avoided, as the target group is broad and not all will identify with those labels. For the second round of testing, the roles were renamed to Practicer and Supporter. This too resulted in confusion especially in the Spanish version, as Practicante is a word that raised more questions than clarity. Back to the drawing board I came up with the names Supporter and Participant. In the last round of testing we had no misunderstandings about user roles making it safe to say that these names were fitting.



Role order during registration

Issue

The order in which the roles were presented was also illogical. Since the user with MCI is the primary focus of the platform, it makes more sense to present this role first. Doing so reduces unnecessary cognitive effort, as participants do not have to read through the supporter explanation before selecting the appropriate option. Placing the most relevant role at the top helps set participants up for success by guiding them toward the correct choice.

Solution

Reorder the roles so that the *Supporter* role is presented second. This better aligns with the expected behaviour of both user groups and reduces the likelihood of participants with MCI selecting the wrong role during registration.

Uncertainty about particpants accounts

Issue
Some testers were unsure whether the person they were supporting needed to already have an account. Although the field is marked as optional, this was not sufficiently clear and caused uncertainty during registration.

Solution
Adding a short explanatory description stating that the supported user does not need to be registered yet. This clarification would reduce confusion and support smoother onboarding.

User profiles title

Issue
The title *User profiles* caused confusion, as it was unclear whose profiles were being referred to.

Solution
Rename the section to a more specific title that clarifies its purpose, such as *People you Support*.

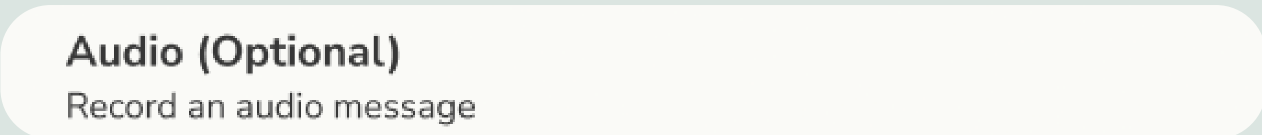
Add supporter not found

Issue
Several testers were unable to find or recognize the Add Supporter button. This is likely due to its position lower on the page, requiring testes to scroll. As a result, the function was overlooked.

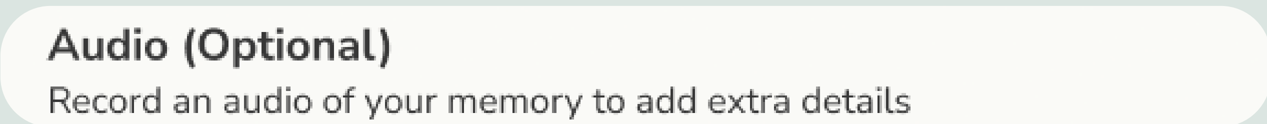
Solution
Move the Add supporter option higher in the settings layout. This also makes more sense in the hierarchy of other settings since adding supporters is quite an important function.

Upload audio

Issue
The description for uploading audio does not clearly explain its added value. As a result, testers tended to add audio by default, even when it did not meaningfully contribute to the memory.



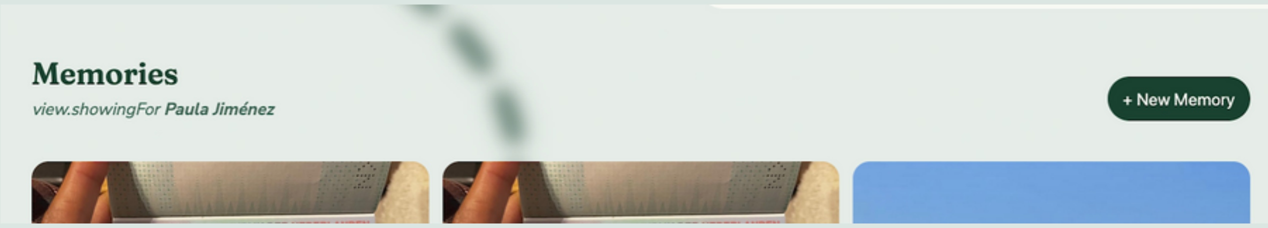
Solution
Improve the description by clearly explaining when and why adding audio is useful. This would help users make more deliberate decisions and prevent unnecessary use of the feature.



Create buttons inside Memories

Issue
Some testers said they were missing a way to create a new memory from the Memories/View page.

Solution
Add a create button inside Memories and inside Practice overview.



Translate option only when logged in

Issue
When users are not logged in there is no option to change the language and the site will be in English by default.

Solution
Having the website take on the language settings of the computer. This way a Spanish computer would show a Spanish site. On top of that I adjusted it so that when any of the other languages in Spain, like Valencian, were the default language it would set to Spanish as well.

Participant Testing

Home button

Issue
Multiple testers experienced confusion around the **Home** button. In Spanish, a *Home* button is translated as *Inicio*, which literally means *start* or *begin*. For testers who are not familiar with technological terminology, this made the button ambiguous. During testing, many navigation-related questions resulted in testers repeatedly returning to *Inicio*, even when it was not relevant to their task.

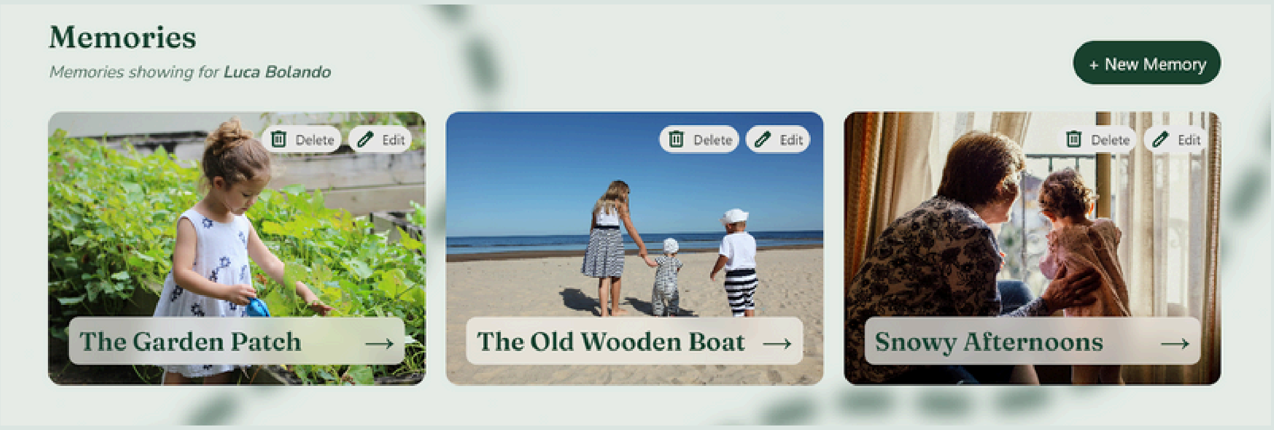
Solution
Because the Home button caused confusion rather than clarity, and the homepage offers limited functional value after login, keeping it in the navbar does not support the user flow.
The Home button will be removed from the navigation bar.
To maintain access to the demonstration video that was previously only available on the homepage, an additional navigation button will be created. During testing, we observed that uncertain participants actively noticed and used the **"view demonstration"** button, while supporters always skipped over it. This indicated that an explanatory video would provide added value for participants who are seeking for more guidance.

Testers also mentioned that once they understood the structure and functionality of the platform, it was very easy to use. Based on this feedback, we conclude that a demonstration video can help participants reach that level of understanding earlier in the process. By placing the link to the video directly in sight, it becomes a clear and intentional support tool.

Memory

Issue
Testers did not realise that the memory element was clickable. Because they felt uncertain, they hesitated to move the mouse around the screen to explore. In addition, the concept of hovering over elements as a way to interact with them was not clear to them. As a result, none of the testers were able to find how to edit a memory, since this required clicking on the image and title to open a modal.

Solution
Place the edit and delete buttons directly on top of the image. This makes it immediately clear that the memory is interactive and that there are actions available beyond simply viewing the image. On top of that, placing an arrow next to the title provides a visual cue that participant can navigate to more details.



Missing text at buttons

Issue
Some buttons lacked explanatory text, such as the record and hint buttons. As a result, testers struggled to understand the purpose of these buttons and were unsure how to use them.

Solution
As stated in my research, buttons should not be ambiguous or leave their function unclear. Both the record and hint buttons will be provided with clear, descriptive text to explain their purpose.

Key findings
Testing with the target group, even though the group was small, provided valuable insights. In particular, the older couple highlighted an important limitation of design: even when an interface is clear and well-structured, it cannot overcome a strong aversion to or fear of technology.

The woman in this couple sat with her arms crossed in front of the computer and when asked what she would do to navigate to a specific part of the platform, repeatedly stated that she never uses a computer or that she would not take any action at all. This behaviour showed a lack of willingness to engage, rather than a lack of clarity in the interface. This reluctance may stem from fear of technology.

From this observation, we conclude that this group of participants will not use the platform independently. This insight not only helps define the realistic boundaries of the platform but also confirms the decision to design the system with separate roles for participants and supporters. This test clarifies that usability improvements alone cannot address all user barriers. By allowing supporters to assist or take over interactions when needed, the platform accommodates participants who are unable or unwilling to engage independently, without excluding them from participation.

One of the testers who visited the hospital participating in the usertest

Conclusion

Answer to the Sub-questions

What is mild cognitive impairment, and how does it affect the way patients interact with digital interfaces?

MCI causes difficulties with memory, attention, decision making, and processing speed. This makes complex interfaces, multi step flows, and dense screens harder to use. Users with MCI are more likely to feel disoriented, forget what they were doing, or hesitate when multiple actions are available. Because the target group consists mostly of older adults, age-related physical decline, digital illiteracy and digital anxiety further increase these challenges.

Why is it important to have the internet be accessible for older people?

Digital platforms have become an unavoidable part of everyday life, and not using digital environments can lead to social exclusion. Age-related physical decline and cognitive impairments, such as difficulties with memory, attention, and processing, can make standard websites hard to use, resulting in loss of independence. Ensuring that the internet is accessible for older adults with mild cognitive impairments is therefore important, particularly as this group is rapidly growing. On top of that, future generations of older adults will have greater experience with digital technology than the current generation. Excluding this demographic group is not only financially unwise but also socially and ethically problematic.

What are the needs of older adults with mild cognitive impairment when interacting with digital interfaces?

Older adults with MCI need interfaces that reduce cognitive load and anxiety. They need to easily understand where they are, what they can do and what will happen when they click something. This means they need systems with clear feedback, confirmations and simple recovery from mistakes. Emotional reassurance is just as important as functional usability.

Which design principles and interaction patterns are most effective for this target group?

The most effective designs use simple layouts, consistent navigation, clear labels and large readable elements. Interfaces should avoid visual clutter, unnecessary animations and abstract terminology. Step by step flows, visible feedback and predictable interactions help participants stay confident.

Answer to the Main Research Question

How can I design and develop a user-friendly interface for a cognitive-stimulation platform that helps older adults with mild cognitive impairments?

A user-friendly interface for older adults with cognitive impairments is achieved by shifting the focus from feature-rich and stimulating interfaces to an experience centred on clarity, guidance and calmness. Rather than overwhelming participants with options, the interface should communicate its purpose and functionality in an intuitive and self-explanatory way.

This means designing interactions that actively reduce uncertainty. The platform should reassure participants by clearly indicating that they are on the right path, provide predictable behaviour, and offer forgiving interactions such as the ability to go back or undo actions without negative consequences. These elements help participants feel safe to explore the platform without fear of making mistakes.

From a development perspective, accessibility must be embedded throughout the entire process instead of being treated as a minimal requirement or final checklist. Structural clarity, consistent interaction patterns, and accessible technical implementation ensure that the interface remains usable and supportive as participants' abilities vary or change over time.

Ultimately, the platform is successful when it minimises cognitive effort and creates a supportive environment in which participants feel capable, oriented and comfortable engaging with memory-based cognitive stimulation activities.

Reflection

Process: Balancing Research, Design, and Development

In my Creative media and Game technologies studies, we are encouraged to follow a structured workflow to minimize uncertainty about whether a product meets the needs of its target group. Normally, this means starting with research, validating the findings, creating the design, testing it and only then beginning development. This prevents ending up with a product that nobody wants or one that still needs major adjustments at the end.

However, this project took a very different approach. To properly test whether the platform truly supports people with MCI, large-scale user testing would be needed, something that is not realistic within the scope of this project. The kind of research required would be highly scientific and belong to a different work field than my colleagues and I practice, so the assumption that the platform is beneficial is largely based on existing literature rather than our own extensive user study. Even so, I had already started my graduation project and the project still needs to be built and tested as effectively as possible within the given constraints.

We had set up an interview with an expert in the field of dementia. For this interview we wanted to show him the project in the shape of a functioning prototype. This meant that after completing the initial UI research, I moved directly into designing and developing a prototype for the upcoming interview. Normally I would prefer to test the design first before moving to the development stage since this could prevent extra work code alterations. However, because the interview with the neurologist was scheduled far into the future due to his limited availability, waiting to begin development until after the interview would have placed the project under a lot of time pressure. In the end the confirmation of the meeting took a long time, and the exact date and location were only known to me one week in advance. So looking back, starting development earlier was the right decision.

For developing the prototype I tried to focus on creating elements that are unlikely to change, typography, spacing, layouts, and components in a reusable way. I set up centralized styles for colors, fonts, and spacing, and build components like buttons as shared modules. This ensured that a lot of updates could be made in one place without duplicating code across files, making the prototype easier to adapt later. This is in general best practice.

By developing and designing simultaneously the whole process started to feel quite messy and out of control at some times.I noticed that trying to work on everything at once, jumping between design, development and writing, created chaos in my head and in my work. Switching constantly between tasks made it difficult to follow my planning and I ended up abandoning it. After the interview with the neurologist and reviewing the first prototype, I realized it was time to get into a calmer and more structured workflow. The plan was to first complete all the designing before looking for feedback in testers. I created a semi-dynamic prototype in Figma, which provided more interactive output than a static design. With this prototype I gathered feedback from the supporter testers group.

After gathering insights, the design was iterated and the necessary changes were implemented. These updates were also applied to the working prototype, with the goal of having a functional version that users with MCI could test. This order was chosen because user testing with people with MCI was expected to take place very late in the project. The interview with the neurologist, who had access to suitable testers, also occurred late in the timeline.

Contact with the neurologist was difficult and confirmation for user testing was only received the night before it was scheduled to take place. Although this was unexpected and made the process chaotic, the situation was manageable because a fully functional prototype was already available.

These last-minute notices required a high level of flexibility and careful prioritisation, particularly in deciding which features needed to be finalised first and which could be postponed.

Completing the user testing with people with MCI brought the research and testing process full circle. At that point, the only remaining step was to implement the final changes into the platform.

Limitations

What didn’t make it

Some features could not be implemented in the final product due to dependency on the delivery of others. The AI model required for these features was only delivered one day before the final usability testing with participants. Implementing these elements afterward would have meant they remained untested and therefore might have been unfit for the target group.
The features that could not be implemented because they depended directly on the AI model are outlined below:

Filters

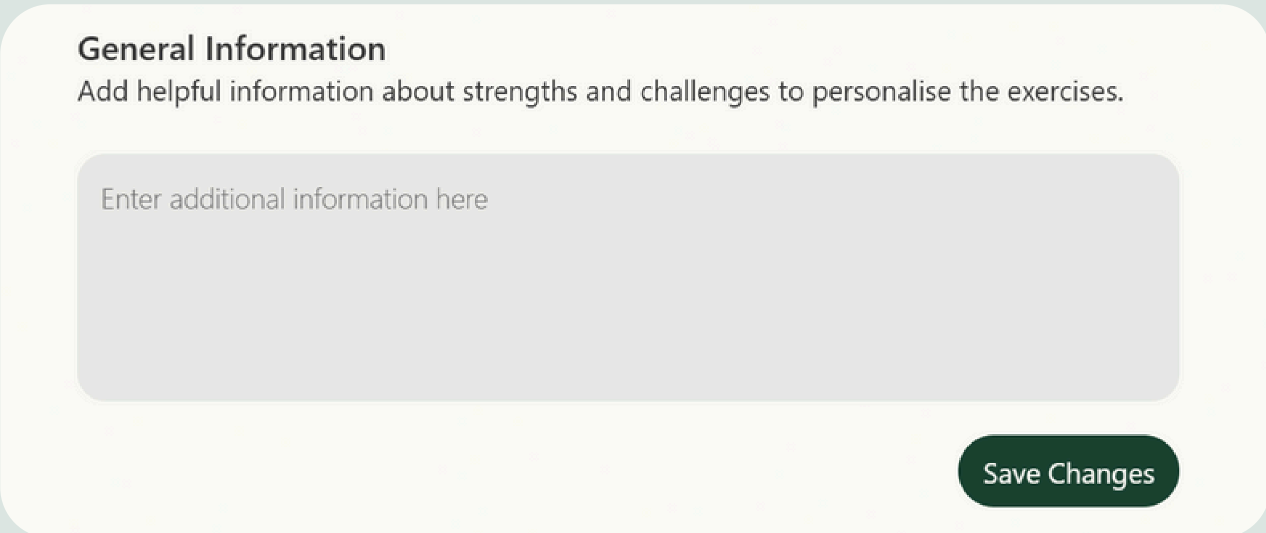
The filter tags of the memories were intended to be automatically assigned by the AI model when a memory is created.

Results

Displaying results required generated exercises and completed exercises first. Since exercises could not be generated without the AI model, results could not be produced. The results view was intended to show the participants progress, highlight which tasks were more difficult or easier, and indicate potential improvements over time.

General information

This feature was intended to allow supporters to provide additional information to the AI model, such as the participant’s strengths and weaknesses, to further personalise the exercises. Without the AI model in place, this information could not be processed and was therefore excluded from the final product.



Use of tools (AI)

For this project, a language model was used in several ways. Primarily, I used it to improve my English writing, as English is not my native language. I first wrote all texts in either English or Dutch and then used the language model to refine wording and readability. All content remains my own; the language model was only used to improve clarity and improve the overall reading experience. I also used a language model to generate images for either decorative reasons or to clarify a point in my paper.

Lastly, a language model was used to help me with setting up and managing the database and the AI model for the exercises. As stated at the beginning of the paper, the primary focus was on the user interface rather than the backend. Since my experience with backend technologies is limited, I used AI to help me navigate this area.

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